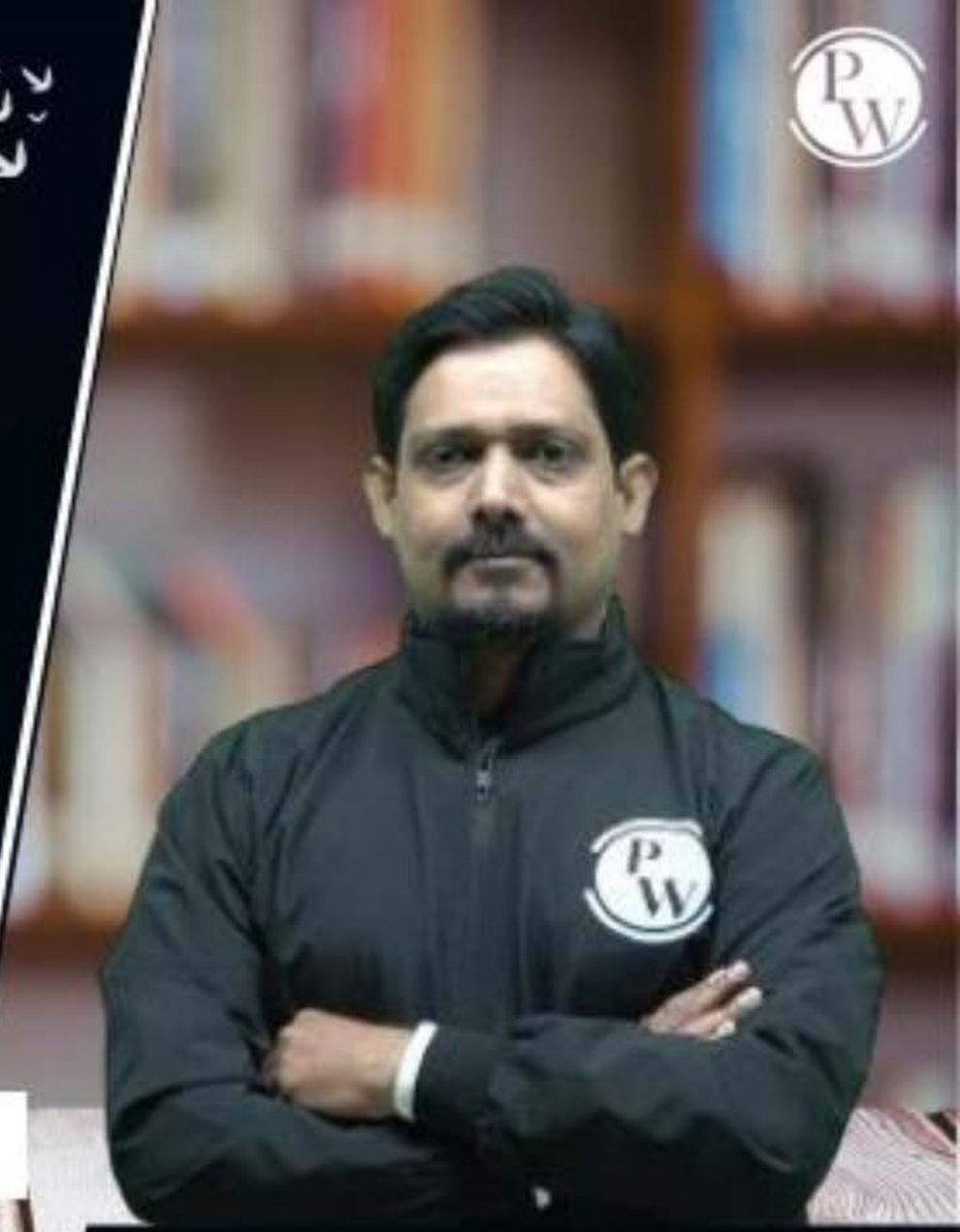
Computer Science & IT



Data Structure & Programming

Linked List

Lecture No. 05



By- Abhishek Sir

Recap of Previous Lecture



- Topic
- Topic
- Topic
- Topic
- Topic

Single Linked

Reverse, middle of list

Insert x after y

Slide

Topics to be Covered



- Topic
- Topic
- Topic
- Topic
- Topic

Doubly linked list

Circular Linked.

Slide



Topic : doubly Linked List

Head pointer



Slide



Topic : Doubly Linked List

Head pointer



Node



extra Space

Browser : forward & backward

Doubly linked

Slide



Topic : Doubly Linked List

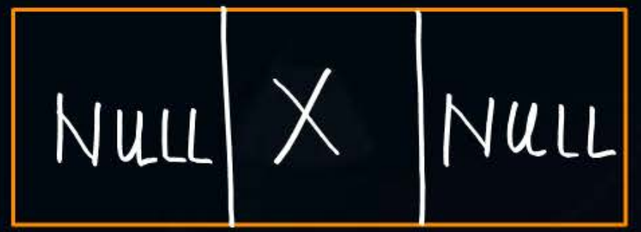
```
typedef struct dnode{
    int data;
    struct dnode * next;
    struct dnode * prev;
} Dnode;
```





Topic : Doubly Linked List

```
Dnode* getnode(x) {
    Dnode *temp = malloc(sizeof(Dnode));
    temp→data = x;
    temp→next = NULL;
    temp→prev = NULL;
    return temp;
}
```



Node

Slide

Topic : Doubly Linked List



HP 100

global variable

temp1 100

temp2 200

temp3 300

HP = temp1;

}

```
void Build123( ) {
  Dnode *temp1, *temp2, *temp3;
  temp1 = getnode(1);
  temp2 = getnode(2);
  temp3 = getnode(3);
  temp1→next = temp2; //①
  temp2→next = temp3; //③
  temp2→prev = temp1; //②
  temp3→prev = temp2; //④
```

Slide

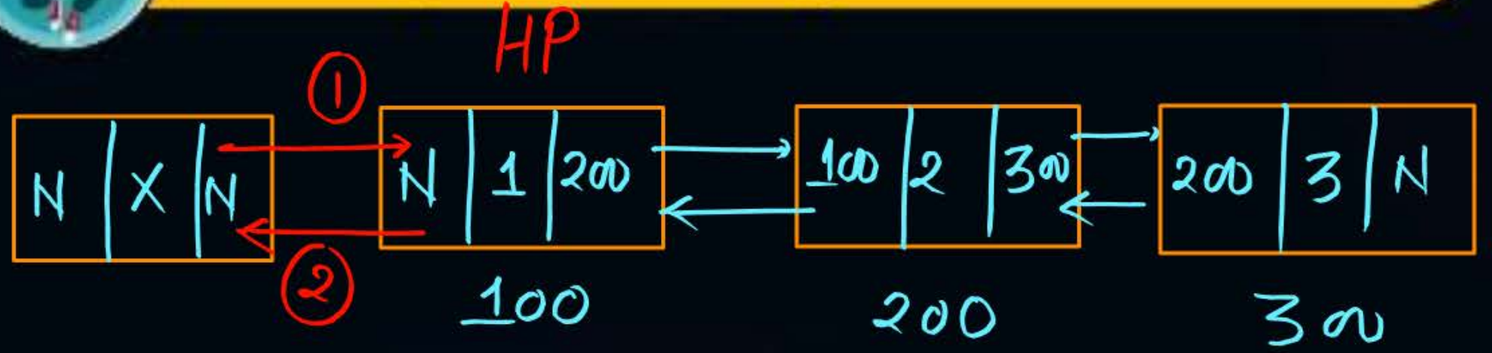
The operation which depends upon length

$O(n)$

The operation which independent of length

$O(1)$ ← constant time.

Topic : Doubly Linked List



p

```
void Addbegin(int x){
Node* p = getnode(x);
if(HP == NULL) // Empty List;
    HP = p;
else{
    p -> next = HP; //①
    HP -> prev = p; //②
    HP = p;
}
}
```

Slide



Topic : Doubly Linked List

Q. Suppose each set represented as a Linked List with elements in arbitrary order. which operation among union, Intersection, membership, cardinality will be Slowest

(a) union only

(b) Intersection & membership

(c) Membership & cardinality.

(d) union, Intersection. —— [D] Multiple traversal Needed

$S = \{1, 2, 3\}$

$1 \in S$? } membership
$4 \in S$? } $O(n)$

Slide



Topic : Doubly Linked List



{ 3, 1, 2 } distinct element.

Cardinality → count while(S != NULL) count++; ✓ | O(n)

Membership → traverse (Linear Search) = O(n)

Intersection 1 → 3 → 2 { 1, 3, 2 } = S_1

$S_1 \cap S_2$

4 → 1 → 5 { 4, 1, 5 } = S_2

1

worst case

1 - 2 - 3

4 - 5 - 6

Slide



Topic : Circular inked List

finding duplicate for every element in S_1 (n) S_2 (n) has to be traverse

n times traverse. $n \times n = O(n^2)$

Union $1 \rightarrow 2 - 3 = S_1$ $S_1 \cup S_2 = \{1,2,3,4\}$

Remove duplicate $3 - 1 - 4 = S_2$

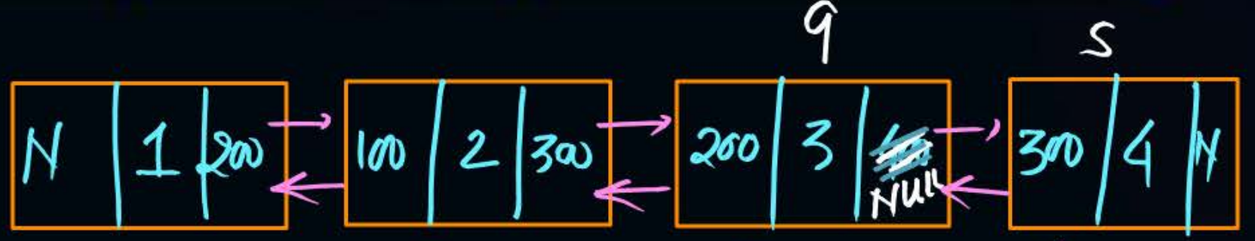
$1 \rightarrow 2 - 3$

$4 - 5 - 6$

n times traversal $n \times n = O(n^2)$

Slide

Topic : Doubly Linked List



if only one Node is present

What problem code will have

Null pointer dereferencing if we directly q→next = NULL

```
void deletend() {
  Dnode *S=HP, *q;
  if(HP== NULL)
      return;
  while(S→next != NULL)
      S=S→next;
  q=S→prev;
  free(S);
  if(q)
      q→next = NULL
  else
      HP=NULL;
}
```

Slide



Topic : Doubly Linked List



S

HP = 100

q = S → prev

q = NULL

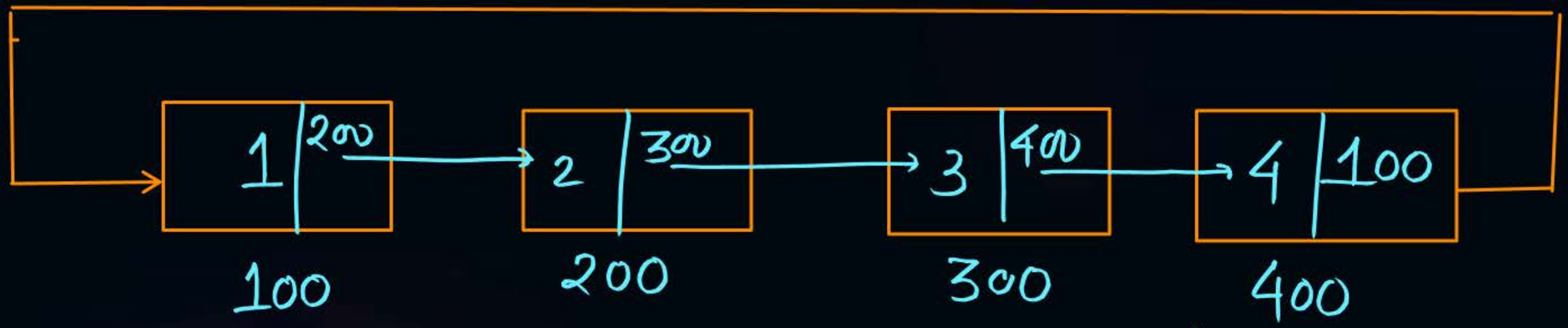
if(NULL)

```
void deletend() {
  Dnode *S=HP, *q;
  if(HP == NULL)
      return;
  while(S→next != NULL)
      S = S→next;
  q = S→prev;
  free(S);
  if(q)
      q→next = NULL
  else
      HP = NULL;
}
```

Slide



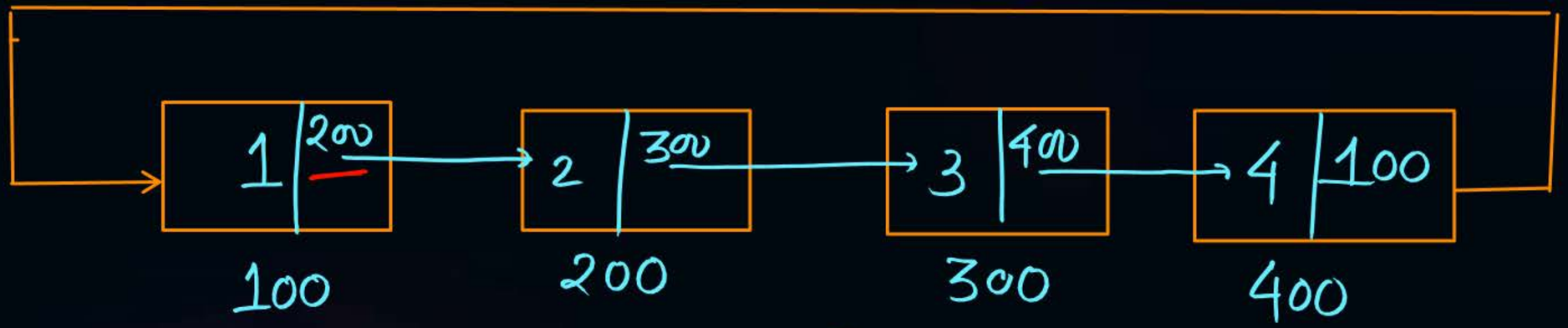
Topic : Circular linked List



* There is No Null terminated Node
* The last Node contains address of first Node forming circle.

Slide

Topic : Circular linked List



HP = 100

Count No. of Nodes

S → next ! = HP
200 ! = 100

Counter 1

S : 200

200 → next ! = 100
300 ! = 100

Counter = 2

S = 300

300 → next ! = 100
400 ! = 100

Counter = 3

S = 400

400 → next
! = 100
100 ! = 100

Slide



Topic : Circular linked List

```
int count() {
    int counter = 0;
    Node *S = HIP
    if(S) {
        while(S->next != HIP){
            S -> S->next
            counter++;
        }
    counter = count+1;
    }
```

return counter;

do while

empty ust ←

Slide



2 mins Summary

Topic

Topic

Topic

Topic

Topic

Doubly Linked List

Circular Linked List

Slide

THANK - YOU